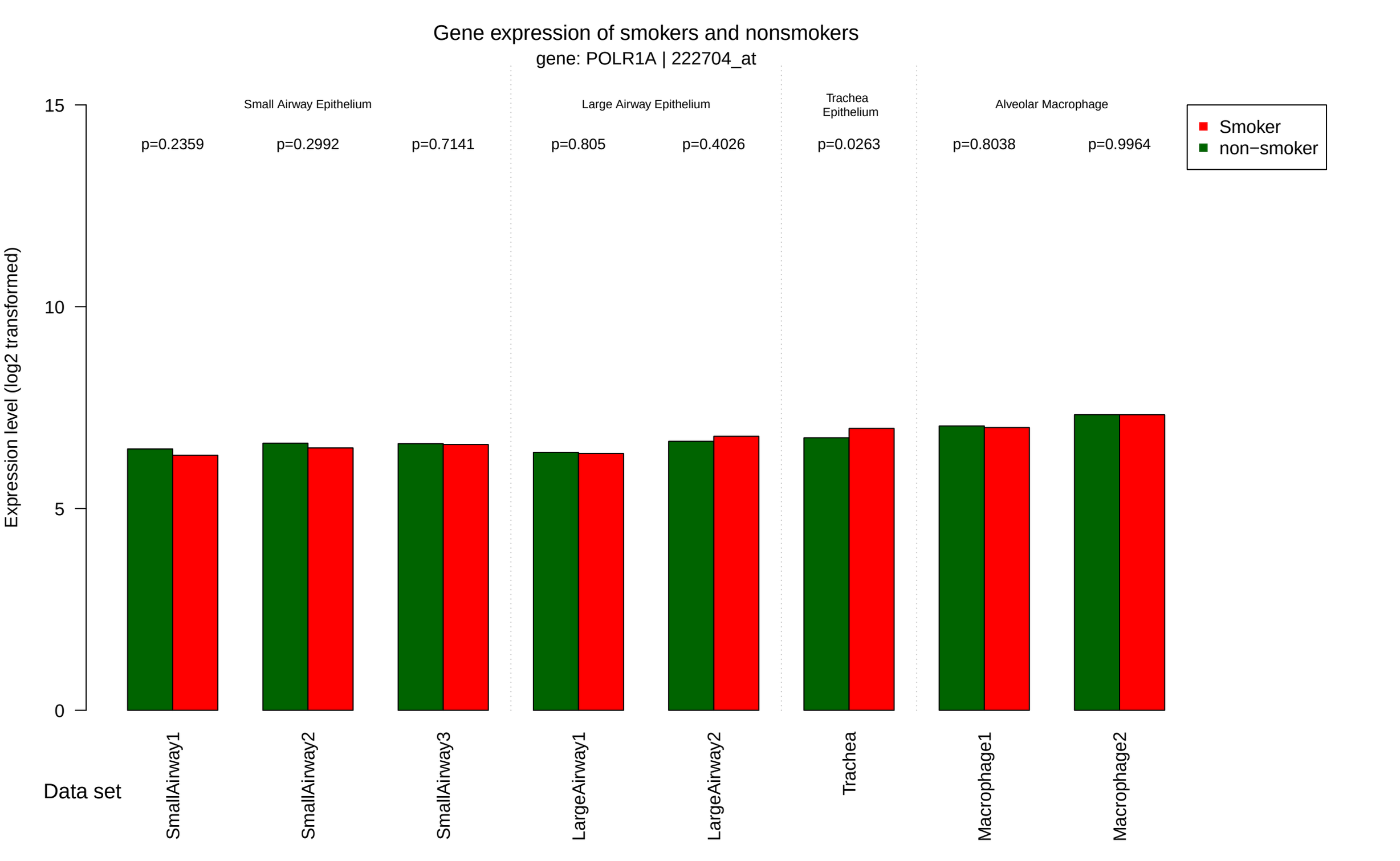
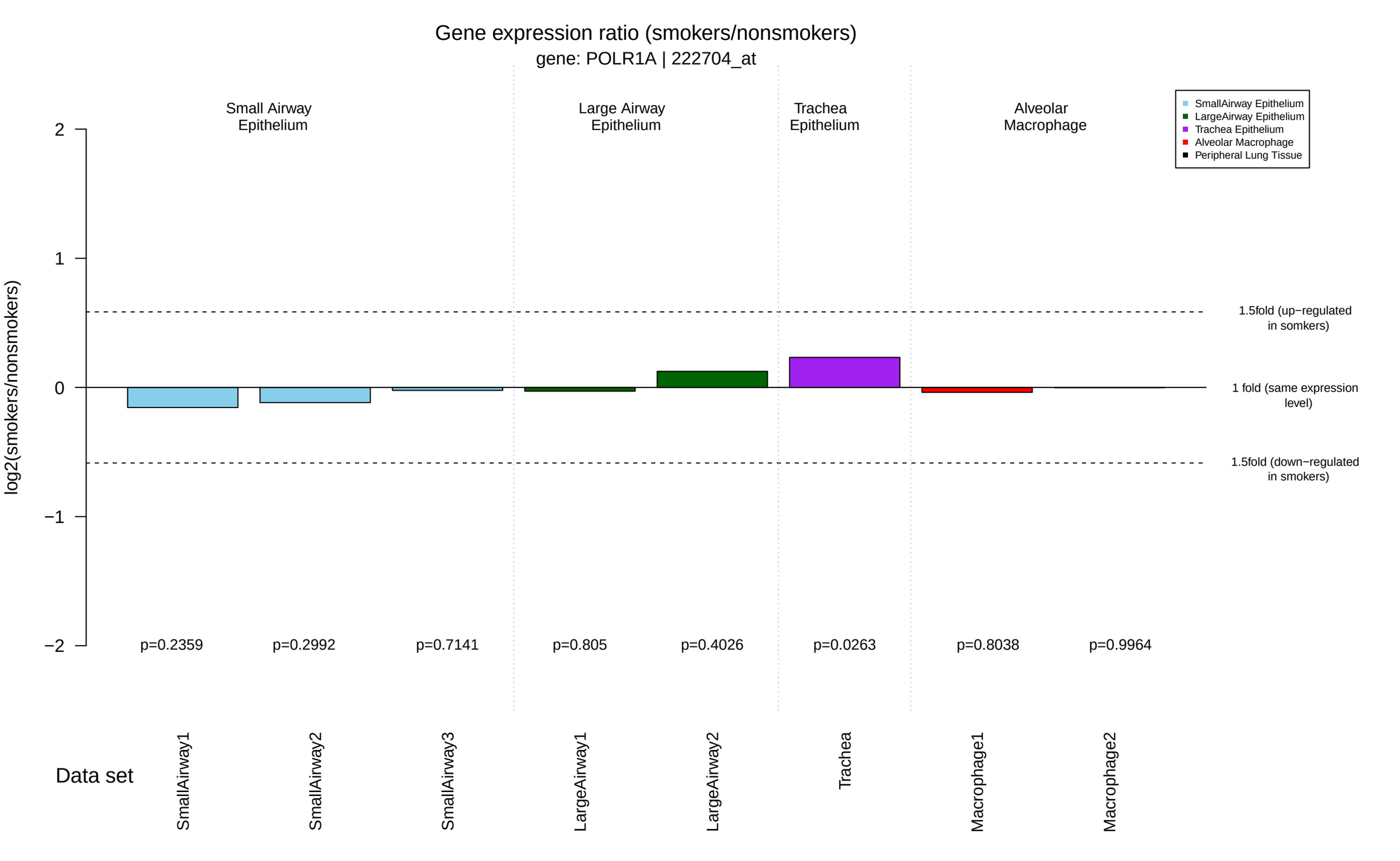
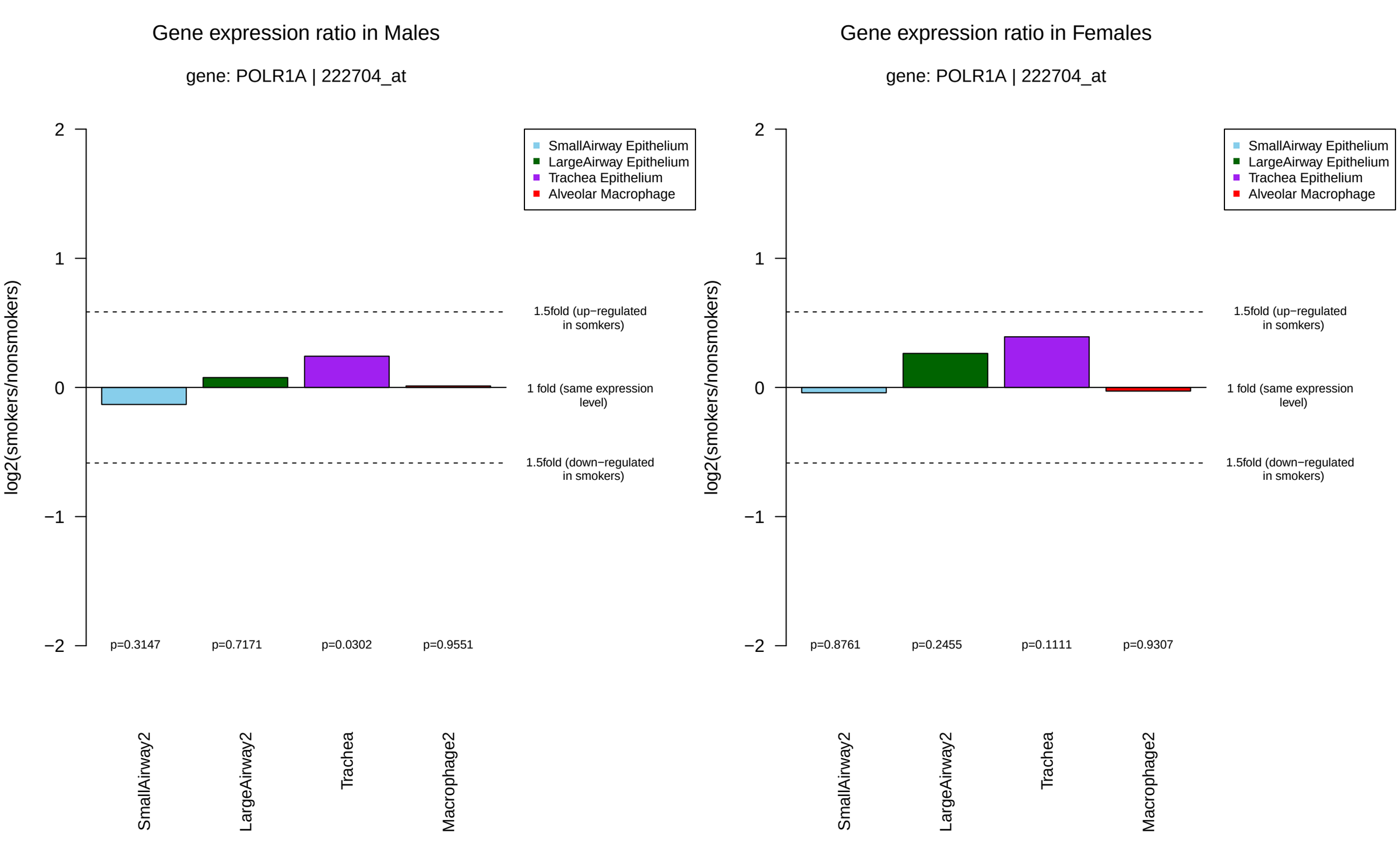
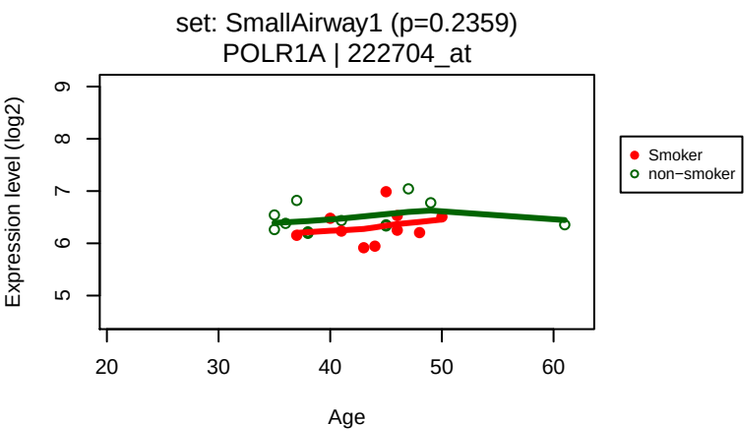
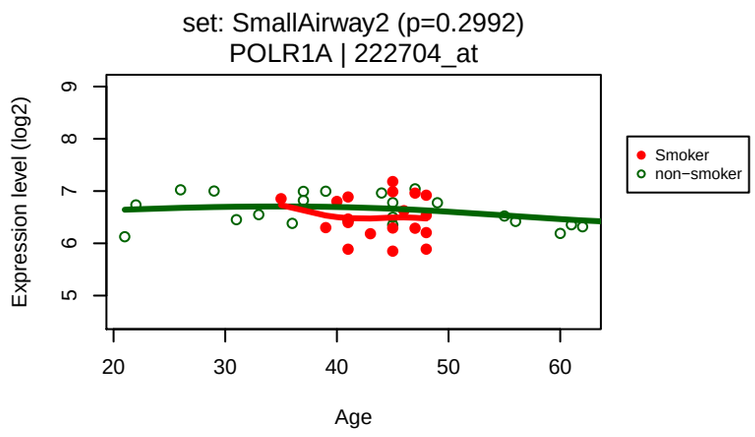
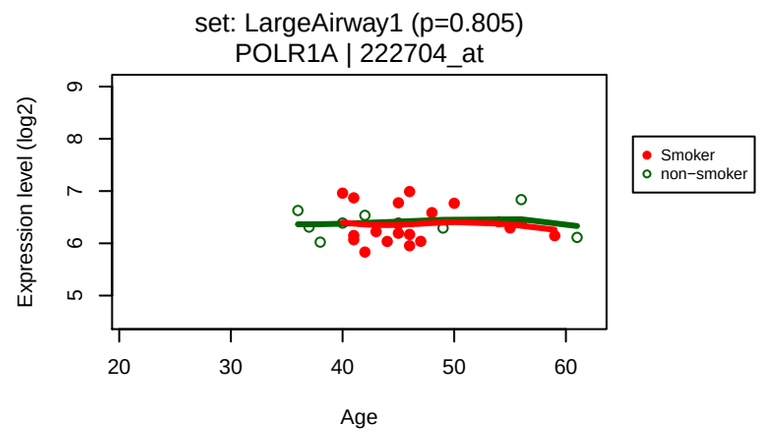

Gene expression of smokers and nonsmokers
gene: POLR1A | 222704_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2359
p=0.2992
p=0.7141
p=0.805
p=0.4026
p=0.0263
p=0.8038
p=0.9964
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

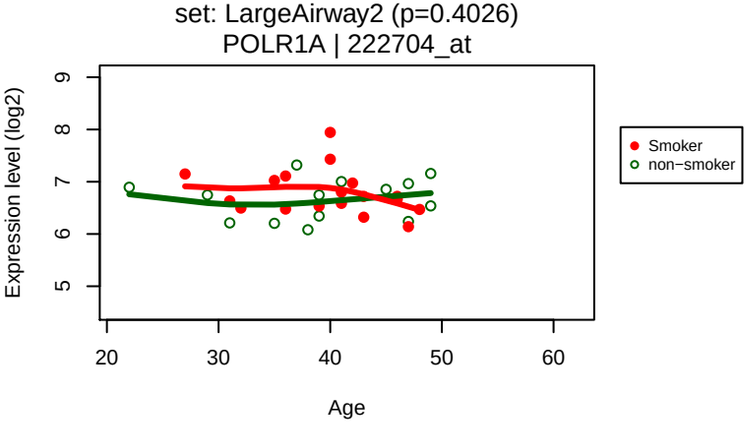

Gene expression ratio (smokers/nonsmokers)
gene: POLR1A | 222704_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2359
p=0.2992
p=0.7141
p=0.805
p=0.4026
p=0.0263
p=0.8038
p=0.9964
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

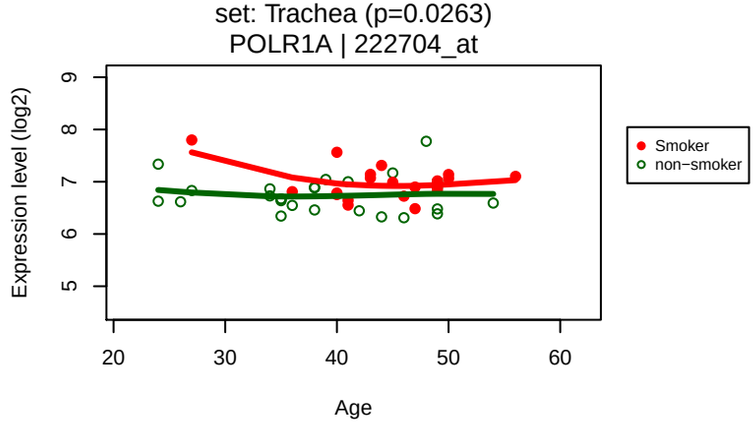

Gene expression ratio in Males
gene: POLR1A | 222704_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3147
p=0.7171
p=0.0302
p=0.9551
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: POLR1A | 222704_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8761
p=0.2455
p=0.1111
p=0.9307
SmallAirway2
LargeAirway2
Trachea
Macrophage2

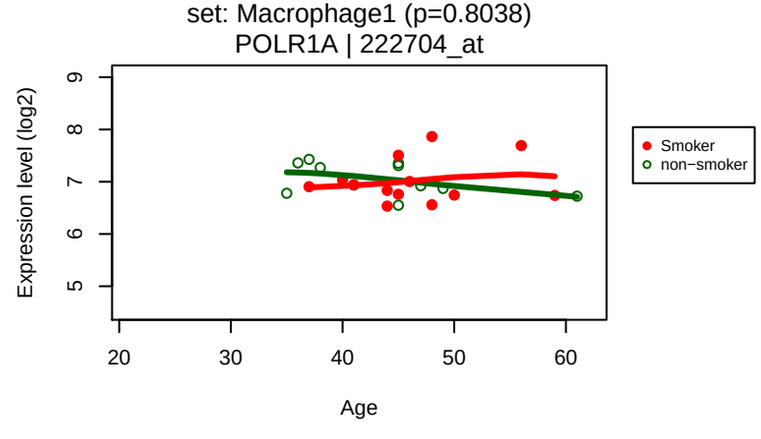

set: SmallAirway1 (p=0.2359)
POLR1A | 222704_at
Expression level (log2)
Age
Smoker
non−smoker

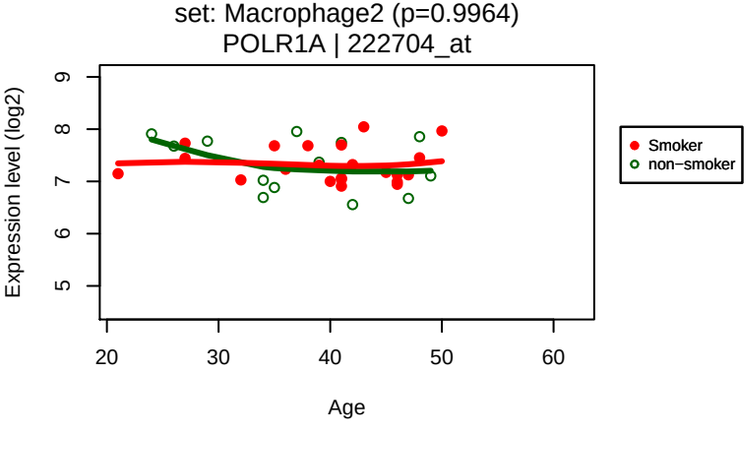

set: SmallAirway2 (p=0.2992)
POLR1A | 222704_at
Expression level (log2)
Age
Smoker
non−smoker

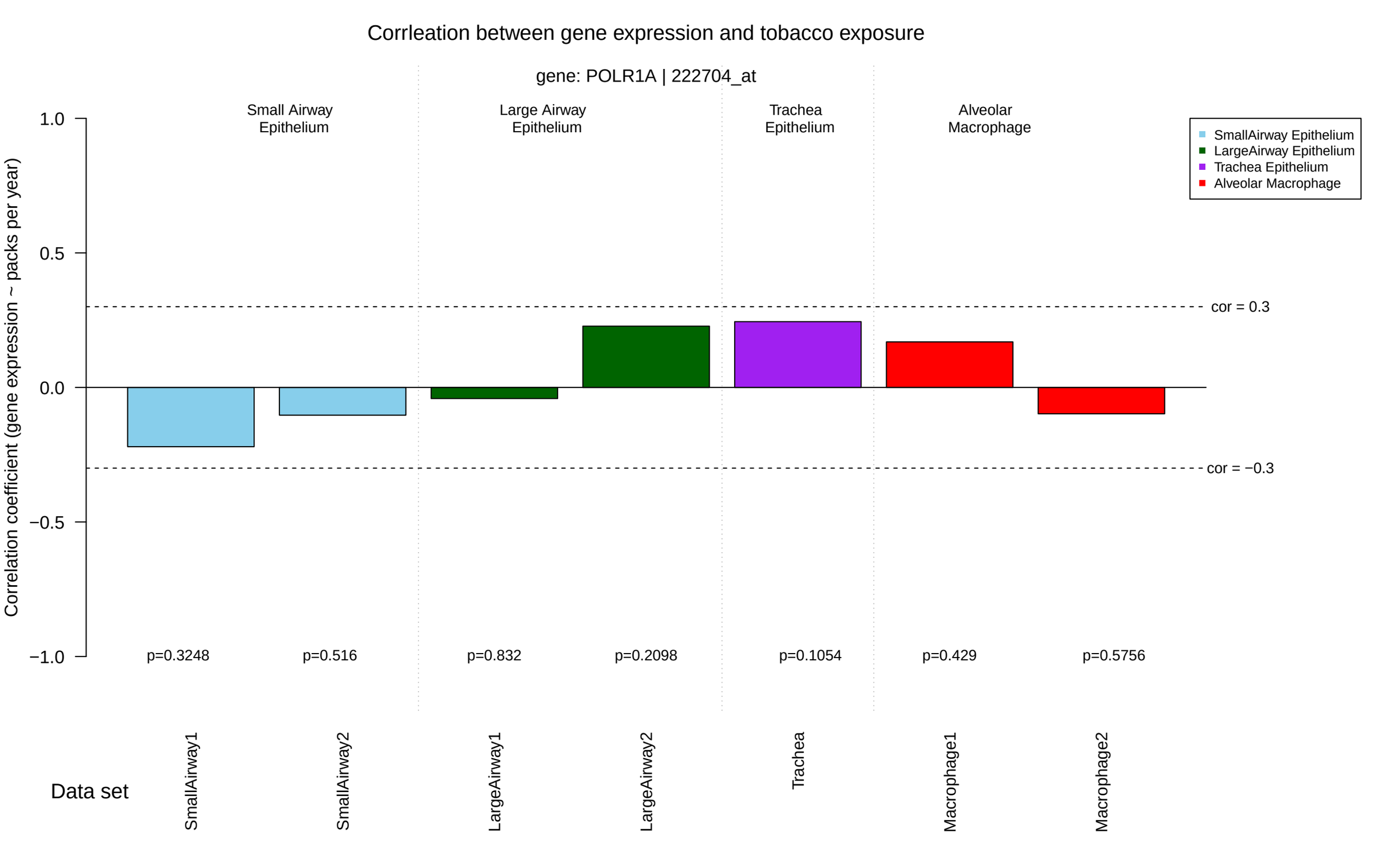

set: LargeAirway1 (p=0.805)
POLR1A | 222704_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4026)
POLR1A | 222704_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0263)
POLR1A | 222704_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8038)
POLR1A | 222704_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9964)
POLR1A | 222704_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: POLR1A | 222704_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3248
p=0.516
p=0.832
p=0.2098
p=0.1054
p=0.429
p=0.5756
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2